

# ***Interim Report – Week 4 Credit Risk Challenge***

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**Project: Credit Risk Probability Model using Alternative Data (Xente eCommerce Platform)**

## **1. Project Understanding**

### **Business Need**

We are building a credit scoring model for a buy-now-pay-later service. The partner eCommerce platform provides transaction-level data but no historical default labels. Our task is to engineer a proxy target variable that identifies high-risk customers and then train a model that outputs a risk probability and a credit score.

### **Basel II & Interpretability**

Basel II requires that models used for regulatory capital calculation (IRB approach) are interpretable and well-documented. Regulators must understand why certain customers are classified as high-risk. A logistic regression scorecard with Weight of Evidence (WoE) transformations is easier to validate than a black-box gradient boosting model, even if the latter may have higher accuracy. Documentation and transparency are non-negotiable in a regulated financial context.

## Proxy Variable Necessity & Risks

Without a direct default flag, we must create a proxy. We will use RFM (Recency, Frequency, Monetary) analysis combined with K-means clustering to label customers with low engagement (low recency, low frequency, low monetary value) as high-risk (proxy for default). Risks:

- **Misclassification** – a creditworthy customer using another payment method could be wrongly flagged high-risk.
- **Proxy decay** – the relationship between RFM and default may change over time (e.g., during economic downturns).
- **Regulatory scrutiny** – the proxy must be justified and monitored.

## Trade-off: Interpretable vs. High-Performance Models

| aspect                | Logistic Regression + WoE     | Gradient Boosting (XGBoost/LightGBM) |
|-----------------------|-------------------------------|--------------------------------------|
| Interpretability      | High – coefficients per bin   | Low – needs post-hoc explanations    |
| Performance           | Good for linear relationships | Higher AUC, handles non-linearity    |
| Regulatory acceptance | Widely accepted               | Often requires extra validation      |
| Development time      | Fast                          | Slower, more hyperparameters         |

In a regulated setting, interpretability often trumps marginal performance gains. We will compare both types and choose the one that balances accuracy and explainability.

## 2. Exploratory Data Analysis – Key Findings

We analysed the Xente transaction dataset (95,662 rows, 16 columns). Below are the key insights, supported by visualisations.

### Data Types & Structure

The dataset contains both numerical and categorical columns:

- Numerical: CountryCode, Amount, Value, PricingStrategy, FraudResult
- Categorical (object): TransactionId, BatchId, AccountId, SubscriptionId, CustomerId, CurrencyCode, ProviderId, ProductId, ProductCategory, ChannelId

· Datetime: TransactionStartTime

```
=== Data Types ===
TransactionId      object
BatchId            object
AccountId          object
SubscriptionId     object
CustomerId         object
CurrencyCode       object
CountryCode        int64
ProviderId         object
ProductId          object
ProductCategory    object
ChannelId          object
Amount            float64
Value             int64
TransactionStartTime  datetime64[ns, UTC]
PricingStrategy     int64
FraudResult        int64
dtype: object

=== Categorical Columns ===
['TransactionId', 'BatchId', 'AccountId', 'SubscriptionId', 'CustomerId', 'CurrencyCode', 'ProviderId', 'ProductId', 'ProductCategory', 'ChannelId']

=== Numerical Columns ===
['CountryCode', 'Amount', 'Value', 'PricingStrategy', 'FraudResult']
```

Figure 1: Data types and column classifications

## Correlation Analysis

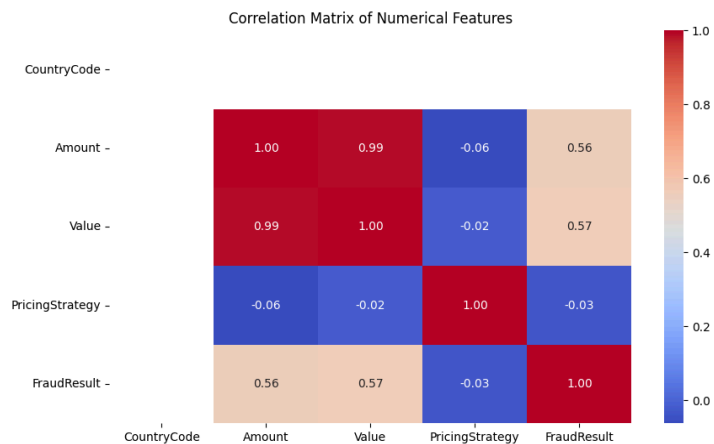


Figure 2: Correlation matrix of numerical features. Amount and Value are highly correlated (0.99). Other correlations are weak, indicating no severe multicollinearity.

## Outlier Detection with Box Plots

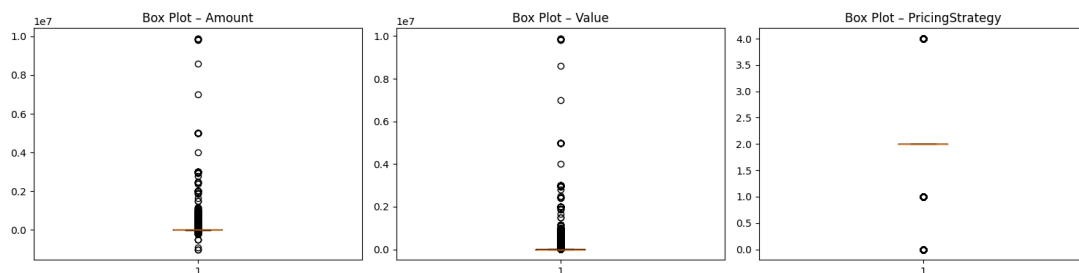


Figure 3: Box plots for Amount, Value, and PricingStrategy. Extreme outliers exist in Amount and Value, requiring transformation or capping.

## Categorical Feature Distributions

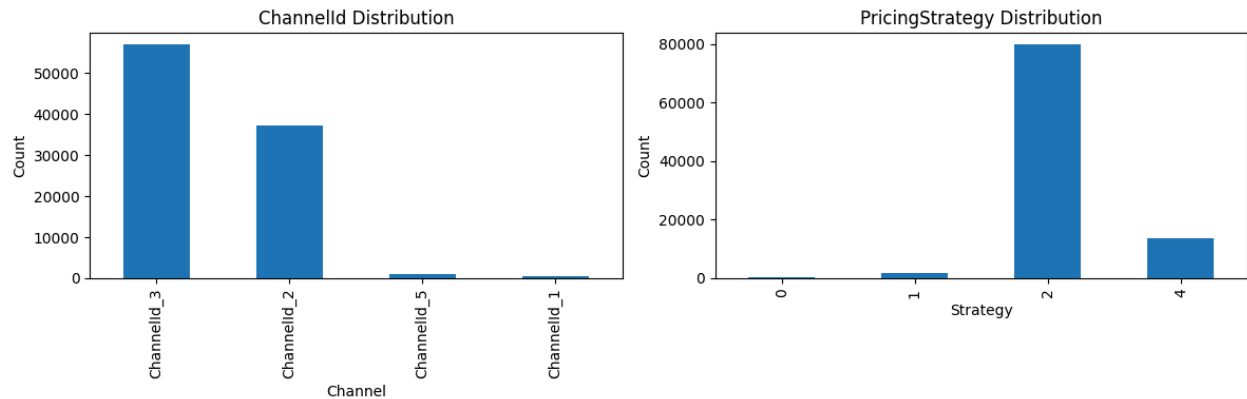
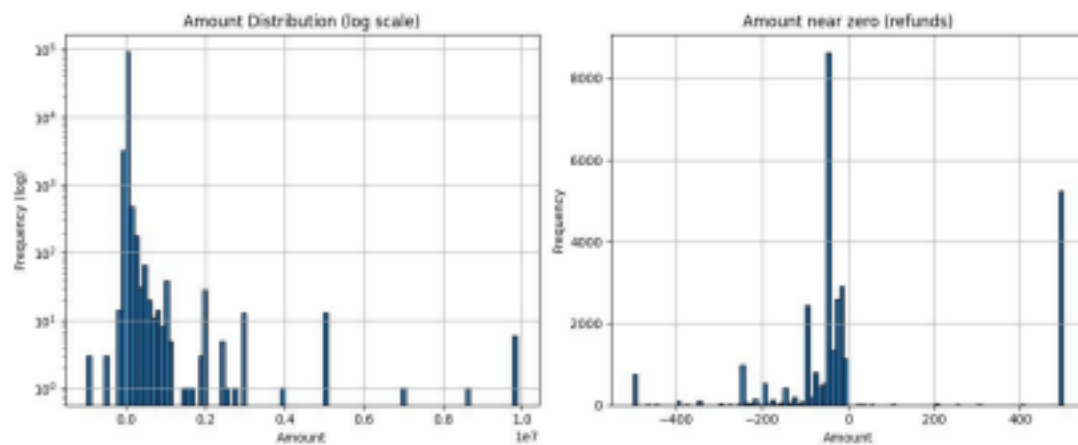


Figure 4: Distribution of ChannelId and PricingStrategy. Channel 2 and 3 dominate; PricingStrategy is concentrated in strategy 2.

## Insight 1 – Transaction Amount Distribution

Transaction amounts are heavily right-skewed – most are small, but there are extreme outliers (up to ~10 million). The log-scale histogram shows that the majority of transactions are below 10,000.

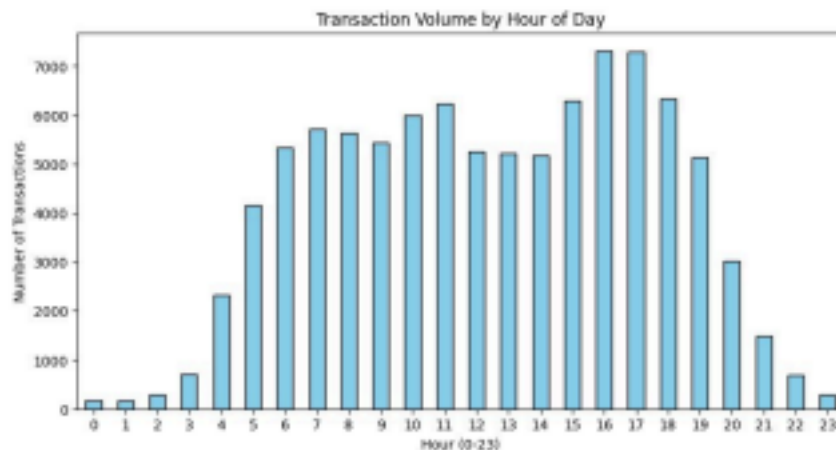


**Business implication:** We will log-transform Amount and treat positive amounts as monetary value for RFM. Negative amounts (refunds) will be handled separately.

## Insight 2 – Transaction Volume by Hour

Transaction volume peaks between 10 AM and 6 PM, with a sharp drop after midnight.

This suggests a daytime-driven platform, possibly with business hours.



**Business implication:** Time-based features (hour of day, day of week) may be useful predictors of customer engagement and default risk.

### Insight 3 – Customer Activity Concentration

Customer activity varies greatly: the top customers drive most transactions (Pareto principle). The customer-level aggregates show a mean frequency of ~25 transactions per customer, but the maximum is 4,091. Monetary spend also has extreme variation (max 83 million).

**Business implication:** This supports RFM-based segmentation. We will cluster customers into 3 groups (high, medium, low engagement) to create the proxy target `is_high_risk`.

### Insight 4 – Negative Amounts (Refunds)

There are negative Amount values (range includes -1,000,000). These likely represent refunds or reversals. The histogram near zero shows a small number of negative values.

**Business implication:** For monetary we will sum only positive amounts (spend). Negative amounts will be ignored for RFM but may be used as a separate feature (e.g., refund count).

### Insight 5 – Missing Values

Missing values are minimal – only a few columns have missing data (<1%). The DataFrame printed an empty missing table, indicating no substantial missing data.

**Business implication:** Simple imputation (mode for categorical, median for numerical) will be sufficient; no need for complex methods.

## Insight 6 – Fraud Analysis (if available)

The fraud rate is 0.20% (2018 fraudulent transactions out of 95,662). Fraud is concentrated in certain product categories (e.g., airtime, digital goods). This may help in feature engineering but is not the target variable for credit risk.



## 3. Next Steps (for Final Submission)

Based on the EDA insights, we will proceed with:

### 1. Feature engineering

- Extract RFM metrics per customer (recency, frequency, monetary).
- Create time-based features (hour, day of week, month).
- Apply Weight of Evidence (WoE) transformation for categorical variables.
- Build a `sklearn.pipeline.Pipeline` for reproducibility.

### 2. Proxy target creation

- Scale RFM features and run K-means clustering (3 clusters).
- Label the least engaged cluster as `is_high_risk = 1`, others as 0.

### 3. Model training & tracking

- Train Logistic Regression, Random Forest, and XGBoost.
- Log all experiments with MLflow.
- Compare models using recall (priority: catch high-risk customers).
- Register the best model in MLflow Model Registry.

### 4. Deployment & CI/CD

- Wrap the model in a FastAPI with a `/predict` endpoint.
- Containerize with Docker and set up a GitHub Actions pipeline for linting and testing.

#### **4. References & Appendix**

- Challenge document: 10 Academy - KAIM 9 - Week 4.pdf
- EDA notebook: notebooks/eda.ipynb
- Dataset: Xente transaction data (95,662 rows, 16 columns)

**End of Interim Report!!!**